

Subword Modeling

11-324/11-824

Spring 2026

For up-to-date information, see the course webpage at
<https://dmort27.github.io/subwordmodeling/>.

1 Instructors

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2 Schedule

Do not panic: students are expected to do **one** reading for each class period (not the whole set of readings).

Table 1:

DATE	TOPIC	READINGS	DUE
Jan 13	Intro Lecture; introduction to Project 1	Haspelmath and Sims (2010) Ch 1–2 Park et al. (2021); Church (2020); Luis Gastaldi et al. (2024)	
Jan 15	Signs, minimal signs, and compositionality		
Jan 20	Productivity and generalization	Haspelmath and Sims (2010); Park et al. (2021), Ch 6	
Jan 22	Inflection, derivation, and compounding	Haspelmath and Sims (2010), Ch. 5; Matthews (1974), Ch. 5; Chaudhary et al. (2020); Hofmann et al. (2021)	
Jan 27	Morphotactics, affix ordering, the mirror principles, and the relevance principle	Aksënova et al. (2016)	

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Table 1: (Continued)

Jan 29	Computational approaches to morphological segmentation/tokenization	Sennrich et al. (2016); Zouhar et al. (2023); Mielke et al. (2021); Creutz and Lagus (2005); Virpioja et al. (2013); Khandagale et al. (2022); Bostrom and Durrett (2020); Hofmann et al. (2022); Limisiewicz et al. (2024); ?	
Feb 03	Allomorphy	Matthews (1974), Ch 6; Haspelmath and Sims (2010), Ch. 10; Yıldız et al. (2019)	
Feb 05	Non-concatenative processes	Amrhein and Sennrich (2021); Klein and Tsarfaty (2020); Fullwood and O'Donnell (2013); Haley and Wilson (2021)	
Feb 10	Lexemes and paradigms; introduction to Project 2	Matthews (1974), Ch. 2; Haspelmath and Sims (2010), Ch. 8	Project 1
Feb 12	Grammatical properties	Matthews (1974), Ch. 9; Sylak-Glassman (2016)	
Feb 17	Rules of realization and rules of referral		
Feb 19	DRM Travel (no class)		
Feb 24	Computational approaches to reinflection and paradigm completion	Neural approaches to paradigm completion	Jin et al. (2020); Wiemer-slage et al. (2022)
Feb 26	Word-and-Paradigm morphology	Matthews (1974), Ch 10	
Mar 3	Spring Break		
Mar 5	Spring Break		
Mar 10	Descriptive phonetics; introduction to Project 3	International Phonetic Association (1999, 1–37)	Project 2
Mar 12	IPA versus orthographies		
Mar 17	G2P and P2G	citetmortensen2018epitran,li2022zero	
Mar 19	Typology of orthographies	Hockett et al. (1997)	
Mar 24	Project Meetings		
Mar 26	Project Meetings		

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Table 1: (Continued)

Mar 31	Unicode—logical and visual representations	(Haralambous and Dürst, 2019)	
Apr 02	Articulatory features; introduction to Project 4	Mortensen et al. (2016), Li et al. (2021), Zouhar et al. (2024)	Project 3
Apr 07	Phonological Similarity	Bharadwaj et al. (2016); Chaudhary et al. (2018)	
Apr 09	Spring Carnival		
Apr 14	Syllabification and syllable segmentation	Bartlett et al. (2009); Mayer (2010)	
Apr 16	Comparative reconstruction	Naik et al. (2025); Lu et al. (2024)	
Apr 21	Project Presentations		
Apr 23	Project Presentations		Project 4

3 Motivation

The goal of this course is to **lead students to engage broadly with the existing NLP and computational linguistics research on subword modeling and develop new computational approaches to problems in morphology, orthography, and phonology**. In addition to three other miniprojects, students will be expected to produce one piece of research that can be developed into a conference or workshop paper (though submission is not a course requirement). The paper should be suitable for the “Phonology, Morphology, and Word Segmentation” tracks of the *ACL conferences, the SIGMORPHON workshop, Coling, or LREC.

Natural Language Processing and Computational Linguistics have traditionally been biased towards phenomenon above the level of the word: syntax, semantics, and discourse.

Contemporary neural models have been very successful in modeling many of these phenomena using words or information-theoretically defined subword units as atomic tokens. However, this has left the structure and patterns that exist inside of words—at the level of morphology, orthography, phonology, and phonetics—relatively unexplored. Of course, there has been computational work in all of these fields for decades but they have never been given the same degree of attention as syntax, semantics, and discourse.

Now, as NLP and CL become increasingly multilingual, interest in languages with richer phonologies, orthographies, and morphologies than English and Chinese has grown. A new wave of research, informed by both linguistic theory and machine learning methods, has opened up novel perspectives on problems in this domain and both practical and scientific issues that have not previously been explored. This course aims to place students on this threshold, reading to make new contributions in this burgeoning subfield.

4 Learning Objectives

At the end of this course, students will:

- Express a sophisticated understanding of the fundamentals of morphology, phonetics, writing systems, and phonology
- Understand current papers in computational phonology, morphology, and orthography
- Implement current subword models of language and carry out publication-quality experiments in this area
- Write publication-quality papers reporting computational research in phonology, morphology, and orthography

5 Mode of Instruction

Class meetings will be structured around handouts (not slides), which will also double as lecture notes (the instructors preferred mode of instruction). This allows a freer and more fluid discussion of the course content than a regimented slide presentation. Furthermore, handouts can provide a rich source of material for student's subsequent reference.

6 Assessments

6.1 Highlights (10 pts)

For 25 of 27 class meetings, each students will submit a list of three highlights: two takeaways from the day's lecture and one question. These highlights are intended to be brief (one sentence each) and to encourage engagement with the lecture. These must be submitted by 11:59pm the day of the corresponding class meeting.

6.2 Projects (60 pts)

The course will be assessed primarily through four mini-projects. Three of the four projects are structured as shared tasks. Students are allowed to work individually or in pairs. All teams are presented with:

- A dataset (train, dev, and test; test labels are held out)
- A metric, an evaluation script, and an autograder for evaluation via Gradescope
- A baseline (usually implementing some of the linguistic insight presented in the lectures in a rule-based model)

Students receive full credit if they beat the baseline on the text set. If they achieve the highest score on the task, they receive 5 bonus points. Unlimited submissions are allowed.

The G2P project is structured differently (see §6.2.3 below).

To receive credit, for each project a student must submit a two page report (ACL template) describing their experiments and the results (as well as referencing earlier work that has inspired theirs).

6.2.1 Morphological Segmentation (15 pts)

dataset Segmented data from two languages (Rarámuri and Shipibo-Konibo).

task Given training and development sets, tag previously unseen words with morpheme boundaries.

metric F1

baselines WFST-based (rule based) segmenter, Unigram tokenization (students must beat both for all languages)

6.2.2 Reinflection and Paradigm Completion (15 pts)

Task 1

dataset Three typologically diverse languages from Unimorph

task Given training and development sets, a previously unseen lemma, and a set of inflectional properties, return the inflected form of the word

metric Exact match accuracy, character error rate

baseline Rule-based word-and-paradigm inflector

Task 2

dataset Three typologically diverse languages from Unimorph

task Given training and development sets and a previously unseen partial paradigm, predict the other wordforms in that paradigm

metric Exact match accuracy, character error rate

baseline Rule-based paradigm completion engine (“rules of referral”)

6.2.3 G2P for Three Languages (15 pts)

Students are required to implement rule-based or data-driven G2P for a language not supported by Epitran2 within the Epitran2 framework (including tests). Students can opt to have their modules included in subsequent uses of Epitran2 and to become coauthors on the Epitran2 paper (in progress).

6.2.4 Cognate Detection (15 pts)

Cognates are words in two or more languages that are descended from the same word in their shared ancestor language (like English *brother* and German *Bruder*).

dataset Mortensen-Wagel Comparative Tangkhulic Database

task In an unsupervised or zero-shot fashion, identify all cognacy relations between words in the four related languages

metric Precision@5 (reference is gold cognacy judgments)

baseline Cosine similarity between mean-pooled phonological feature vectors with empirically-determined threshold

6.3 Final Presentation and Paper (30 pts)

Students will select one of their projects (or a different idea of their choosing), develop it into a potentially publishable research product and make a conference-style oral presentation of it to the class. This presentation will be evaluated both on its technical and scientific merit and on its communicative effectiveness. The same research will be written up as four- to eight-page research paper which will be evaluated generously based on its likelihood to be accepted by one of the target venues.

7 Grading

Grades will be assigned as follows:

ASSESSMENT	POINTS
Highlights	10
Mini-Projects	60
Final Project	30

The grading scale will be as follows:

Grade	Range	
A+	100%	to 94.0%
A	< 94.0%	to 90.0%
A-	< 90.0%	to 87.0%
B+	< 87.0%	to 84.0%
B	< 84.0%	to 80.0%
B-	< 80.0%	to 77.0%
C+	< 77.0%	to 74.0%
C	< 74.0%	to 70.0%
C-	< 70.0%	to 67.0%
D+	< 67.0%	to 64.0%
D	< 64.0%	to 61.0%
F	< 61.0%	to 0.0%

Highlights will be based on reasonable completion. Mini-projects will be based on meeting the base line for each challenge. Scores below the baseline will be credited according to a linear function. The student with the highest score on each task will receive 5 bonus points. The final paper will be based on one of the four mini-projects. The deliverable will be a paper and presentation, which will be graded together according to the following criteria:

1. Is related work handled appropriately? (10%)
2. Is the methodology well-motivated and technically sound? (20%)
3. Are appropriate baselines employed? (15%)
4. Are the experiments well designed? (15%)
5. Does the analysis of the results follow accepted statistical practices? (10%)
6. Is the paper well-written? (15%)
7. Was the structure and presentation of the talk clear and effective (15%)

8 Policies

8.1 Academic integrity

Any cheating or plagiarism will be dealt with according to the University policies on academic integrity. In general, discussion of tools, concepts, and formalisms is acceptable collaboration and is encouraged. Misrepresenting the work of others as your own, however, is considered cheating. You are allowed to use generative AI tools to assist you in writing and debugging code. This is now standard practice in software development. However, you are responsible for all code you submit (and for any unattributed similarity between your code and code that is publicly available). You

are not allowed to use generative AI tools for ideation (only for the mechanical aspects of coding). If you use generative AI tools, you must state in your report which tools you used as how you used them.

8.2 Late Policy

Late work is not accepted except with an accommodation from the Office of Disability Resources, following university policy. If you have experienced an adverse event (for example, an illness) we encourage you to work with ODR to obtain an accommodation, which we will happily accept.

8.3 Non-Discrimination Policy

Throughout human history, some people have been denied the rights and opportunities available to others on the basis of their race, gender, economic class, caste, ancestry, language community, age, religion, beliefs, political affiliation, and abilities (visible and invisible). A single course cannot undo the injustices of history, but we—as a teaching staff—are committed to fighting discrimination. We encourage you to join us.

8.4 Disability Rights

Many people have disabilities, including members of our own families. We see disabilities as deficits not in disabled people but in the institutions and societies that are structured such that they are disadvantaged. We wish to do our part to overcome this disparate treatment. If you have a disability (visible or invisible), please let us know as soon as possible (you don't need to tell us the nature of the disability) and work with Disability Service to develop a set of accommodations which we can then approve. These might, for example, include lecture materials that are usable by people with visual disabilities, sign language interpretation, captioning, flexible due dates, etc.

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